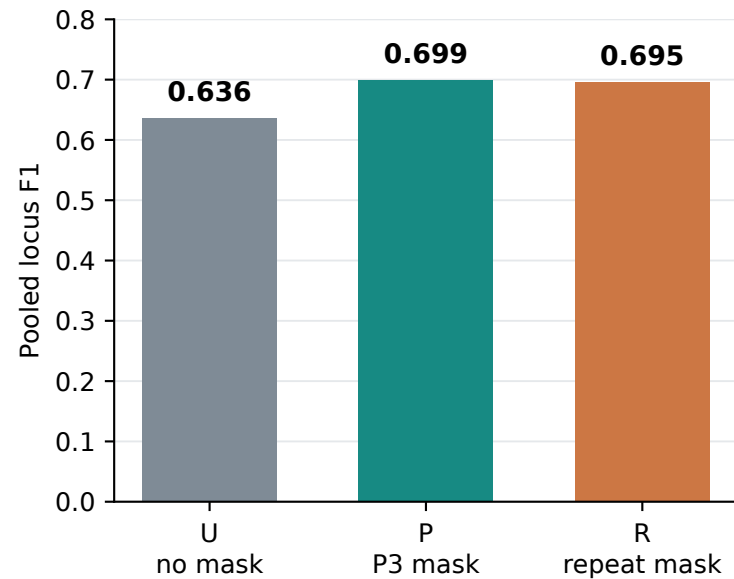
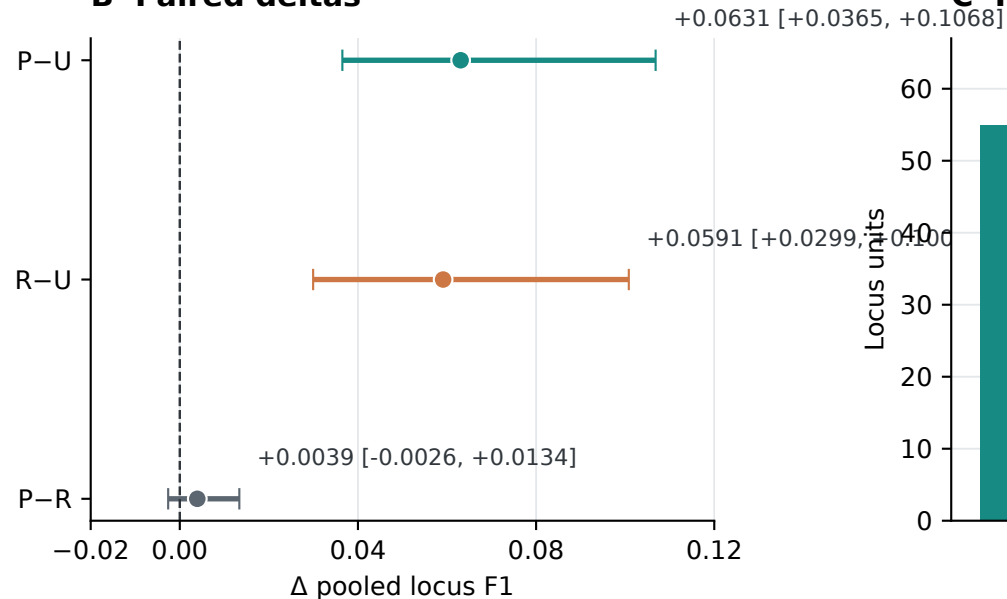


Average input utility improves, but the four-gate criterion is not met

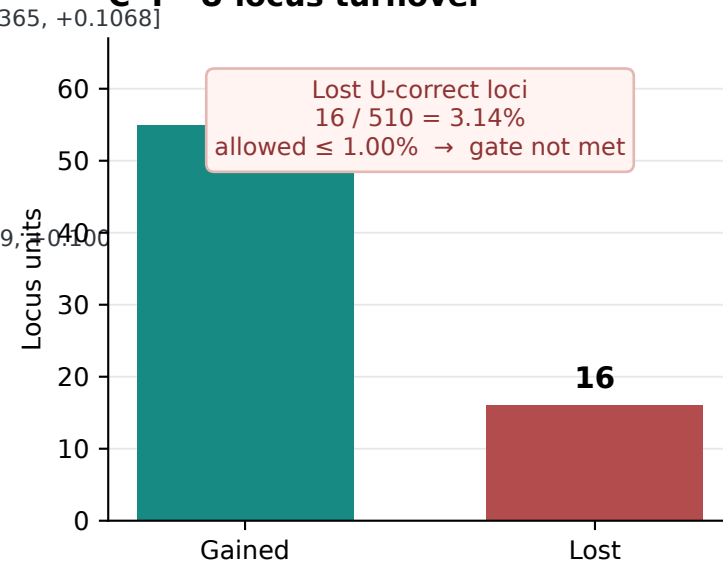
A Pooled locus F1



B Paired deltas



C P-U locus turnover



Fixed hg38 chr16/18 panel · 726 reference locus units · 20 cores $\times$ 3 arms · 60 completed cells.

P-U is prespecified: $\Delta = +0.0631$, 95% CI [+0.0365, +0.1068]; P-R CI crosses 0.

No equivalence or P-over-R superiority claim. Exploratory fixed-panel input utility, not independent generalization or no-risk evidence.